

Owen Fox

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EDUCATION

Villanova University

Bachelor of Science in Computer Science

- **GPA:** 3.72 / 4.00; **Honors:** Dean's List (6 Semesters)

Villanova, PA

Expected May 2026

PROJECTS

Bullpen Engine | *TypeScript, Next.js, PostgreSQL, Prisma, Docker*

- Developed full-stack TypeScript application with Next.js App Router, implementing RESTful APIs for real-time simulation execution and PostgreSQL database integration via Prisma ORM
- Engineered pitcher performance model sampling K/9, BB/9, HR/9 rates to generate plate appearance outcomes, advancing baserunners through statistically accurate win probability calculations
- Containerized application with Docker Compose for single-command deployment, including database seeding with 30 MLB reliever profiles and automated schema migration

PokeMahn | *Godot, C#*

- Developed PokeMahn, a Pokémon-inspired RPG built in the Godot game engine with C#, featuring a tilemap overworld, animated player movement, encounter triggers, and battle logic
- Implemented core gameplay mechanics, including map navigation, collision detection, and encounter systems to simulate exploration and turn-based battle triggers.
- Structured game logic using Godot's scene and node system, modularizing components for environment, player control, UI messaging, and battle events.

Space Invaders | *Python*

- Developed a fully interactive Space Invaders game in Python using the Turtle graphics library, implementing real-time rendering, keyboard input handling, and a custom game loop running around 60 FPS.
- Managed game state and HUD rendering, tracking score, game-over conditions, and user feedback through on-screen updates and event-driven callbacks.
- Designed modular game systems including player movement, projectile mechanics with cooldown logic, alien formation movement with edge bouncing and timed descent, and win/lose state management.

EXPERIENCE AND LEADERSHIP

Vice President

Villanova Software Engineers

Villanova, PA

October 2024 – Present

- Serve as Vice President of a 20+ member student engineering organization building full-stack applications for startup clients, leading frontend development and coordinating closely with backend, design, and client-facing teams
- Actively contributing to the design and frontend implementation of the VIRS Writing Challenge, building start/stop timer persistence, EST-based daily resets, streak validation rules, and role-based UI flows using React, FastAPI, Firebase Authentication, and PostgreSQL
- Mentor 10+ junior members through weekly code reviews and collaborative projects, reinforcing best practices in Git workflows, debugging, and frontend architecture to improve code quality and accelerate onboarding

Competitive Programmer

Villanova Programming Team

Villanova, PA

August 2022 – Present

- Selected to represent Villanova in the upcoming 2026 Swarthmore programming contest, solving complex algorithmic problems in Python under timed conditions and strengthening data structures and algorithmic design expertise
- Solved 200+ algorithmic challenges on LeetCode and Kattis through weekly practice sessions, strengthening competitive programming performance and advancing mastery in algorithmic strategies, dynamic programming, and complexity analysis
- Collaborated in a 3-member programming team to strategize and optimize real-time problem-solving workflows for competitive events, improving coordination, time management, and debugging efficiency across high-pressure contests

Python Pro Bootcamp - 100 Days of Code

Udemy

Remote

February 2024 - May 2024

- Gained proficiency in Python syntax, control structures, and core data structures (lists, dictionaries, tuples, and sets) to build and analyze complex applications
- Developed a solid understanding of object-oriented programming principles in Python, enhancing code reusability and efficiency
- Demonstrated the ability to work independently on complex programming challenges, enhancing problem-solving skills and technical creativity

TECHNICAL SKILLS & CERTIFICATIONS

Frameworks & Languages: JavaScript, Python, Java, React, SwiftUI, TailwindCSS, Vite

Cloud, Tools & Platforms: GitHub, Jira, Vercel, Xcode

Design & Supporting Technologies: HTML, CSS

Certifications: LinkedIn Python and Java Skill Assessments